Curriculum Vitae
Timothy G. Raben

Professional Appointments

Post Doctoral Researcher – Michigan St. Univ. Dept. of Physics&Astronomy	Fall 2018 – present
Post Doctoral Researcher – Univ. of Kansas, Dept. of Physics&Astronomy	Fall 2016 – Fall 2018
Post Doctoral Research Fellow – Brown Univ., Dept. of Physics	Spring 2016 – Fall 2016
Visiting Assistant Professor – Brown Univ., Dept. of Physics	Spring 2016

Education

Brown University, Ph.D. 2016
High Energy Theory Group: adviser—Prof. Chung-I Tan
Brown University, Sc.M. 2012
University of Michigan, 2009, Graduated with Honors
Double Major: Mathematics (Honors), Physics: adviser—Prof. L. Pando-Zayas
Calvert Hall College High School 2005

Papers, Proceedings, and Publications

- *Radial Lattice Quantization of 3D ϕ^4 Field Theory* arXiv:2006.15636
- *Within-Family Validation of Polygenic Risk Scores and Complex Trait Prediction* *Sci. Rep.* bioRχiv:976654
- *Genetic Architecture of Complex Traits and Disease Risk Predictors* *Sci. Rep, 10, 12055(2020)* bioRχiv:946608
- *Genomic Prediction of 16 Complex Disease Risks Including Heart Attack, Diabetes, Breast and Prostate Cancer* *Sci. Rep. 9, 15286(2019)* bioRχiv:506600
- *Lattice ϕ^4 Field Theory on Riemann Manifolds: Numerical Tests for the 2-d Ising CFT on S^2* *Phys. Rev. D98, 014502 (2018)* arXiv:1803.08512
- *Minkowski Conformal Blocks and the Regge Limit for SYK-like Models* *Phys. Rev. D98, 086009 (2018)* arXiv:1801.04208
- *Pomeron Physics at the LHC* Proceedings arXiv:1711.07514
- *Holographic Inclusive Central Particle Production at the LHC* Proceedings arXiv:1711.02727
- *Inclusive Production Through AdS/CFT* *JHEP 11 (2017) 075* arXiv:1702.05502
- *Lattice Dirac Fermions on a Simplicial Riemannian Manifold* *Phys. Rev. D95 (2017) no.11, 114510* arXiv:1610.08587
- *Holographic Approach to Deep Inelastic Scattering at Small-x at High Energy* Proceedings arXiv:1508.05063
- *Towards holographic QCD: AdS/CFT, confinement deformation, and DIS at small-x* Proceedings arXiv:1412.3443
- *Strong Coupling Expansion for the Conformal Pomeron/Odderon Trajectories* JHEP 1502 (2015) 104 arXiv:1409.2730
- *Conformal Pomeron and Odderon in Strong Coupling* Proceedings arXiv:1312.1419

Funding: PI for an N.I.H. K-25 grant application. Co-author on two D.O.E. grants (one successfully awarded $400k), two N.S.F. grants, a D.O.D. grant, and a N.A.S.A. grant.

Talks

"Looking for Pomerons at the LHC"	*MSU HEP seminar*	Oct 2018
"Crossing the bridge from BFKL to saturation"	*Diff/Low-x* – R.C. Italy	Aug 2018

"Progress toward identifying BFKL effects in dijet processes"
April APS mtg. – Ohio USA Apr 2018
"Progress toward identifying BFKL effects in dijet processes"
Forward Physics and Diffraction – Univ. Aut. Madrid Spain Mar 2018
"Pomeron Physics at Colliders" *Particle Physics on the Plains*—Univ. of Kansas USA Sept 2017
"Toward Mueller-Tang Jets at Next-to-Leading Order"
International Symposium on Multiparticle Dynamics—Tlaxcala, Tlaxcala Mexico Sept 2017
"CFT at Strong Coupling: from DIS to Inclusive Production"
Low-x 2017—Bisceglie, Puglia, Italy June 2017
"Low-x DIS: Holographic Confinement"
Low-x 2015—Sandomierz, Poland Sept 2015
"AdS/CFT, Confinement, and DIS at Small-x"
Quarks and Nuclear Physics—Universidad Tecnica Federico Santa Maria, Chile Mar 2015
"Regge Physics and The AdS/CFT // Lattice CFT"
Humboldt/Brown Mini-Conference –Brown Univ. USA Oct 2014
"Towards Holographic QCD—AdS/CFT, Confinement, and DIS at Small-x"
QCD and High Energy Interactions—Rencontres de Moriond, La Thuile Italy Mar 2014

Conferences & Visits

"Diffraction & Low-x 2018"(organizing committee & session organizer)Reggio Calabia, Italy Aug 2018
"15th Workshop on Non-perturbative QCD" (session organizer) Paris France June 2018
"April APS Meeting" Columbus Ohio, USA Apr 2018
"Forward Physics and Diffraction" Univ. Aut. Madrid, Spain Mar 2018
"Particle Physics on the Plains" Univ. of Kansas, Lawrence, KS USA Sept 2017
"International Symposium on Multiparticle Dynamics" Tlaxcala, Tlaxcala Mexico Sept 2017
"Low-x 2017" Bisceglie, Puglia, Italy June 2017
"APS April Meeting 2017: Quarks 2 Cosmos" Washington, D.C. USA Jan 2017
"PhotoNuclear Reactions" Gordon Research Conference, Holderness NH USA Aug 2016
"Seventh New England String Meeting" Brown Univ., Providence RI USA Nov 2015
"Low-x Meeting 2015" Sandomierz Poland Sept 2015
"Black Holes, Holography, and Strings: A Celebration of the Science of Andrew Strominger"
(AndyFest) Harvard Univ., Cambridge MA USA July 2015
"7th International Conference on Quarks and Nuclear Physics"
Universidad Tecnica Federico Santa Maria, Chile Mar 2015
"Sixth New England String Meeting" Brown Univ., Providence RI USA Oct 2014
"Humboldt/Brown Mini-Conference" Brown Univ., Providence RI USA Oct 2014
"Strings 2014" Institute for Advanced Study, Princeton Univ., Princeton NJ USA June 2014
Short Term Visitor CERN Theory Division, Geneva Switzerland Apr 2014
"QCD and High Energy Interactions" Rencontres De Moriond, La Thuile Italy Mar 2014
"From Gravity to Strong Coupling Physics"
Yukawa Institute for Theoretical Physics, Kyoto Univ., Kyoto Japan Oct 2012
"Fifth New England String Meeting" Brown Univ., Providence RI USA Nov 2011
"2011 Meeting of the Division of Particles and Fields of American Physical Society"
DPF, Brown Univ., Providence RI USA Aug 2011

Workshops & Schools

"Selected Topics in Holography" Mathematica Summer School on Theoretical Physics
Porto & Lisbon, Portugal July 2014
"Strings Theory" Prospects in Theoretical Physics, Institute for Advanced Study
Princeton Univ. June 2014

"Gauge/Gravity Duality" Long-Term Workshop.
Yukawa Institute for Theoretical Physics, Kyoto Univ. Oct 2012

Programming

Languages: Mathematica Julia Python Bash tex C++/C HTML Fortran
Methods: Finite Elements Lattice Monte Carlo Event Monte Carlo (Pythia, Herwig)
Monte Carlo Integration Machine Learning LASSO Linear Analysis

Awards & Fellowships (Non-Teaching)

Beyer Award—For excellence in scholarship and service Brown Univ. Physics 2016
Brown Faculty Fellowship—Full funding for the final semester of graduate work and appointment as a visiting assistant professor for a semester immediately following graduation 2015-2016
Brown Conference Travel—$650 Travel Grant Aug 2015
Brown International Travel Fund—$600 Travel Grant Aug 2015
Brown Conference Travel—$650 Travel Grant Mar 2015
Brown International Travel Fund—$800 Travel Grant Mar 2015
Joukowsky Summer Research Travel Award—$2000 Summer 2014
Brown Conference Travel—$500 Travel Grant Mar 2014
Brown International Travel Fund—$550 Travel Grant Mar 2014
Golden Key member
Addison-Wesley Award: University of Michigan Physics—Presented to a graduating senior for outstanding contributions to the life of the Physics Department 2010
University Honors—Univ. of Michigan 04/26/2007 12/20/2007 12/18/2008
Sigma Pi Sigma member

Awards & Fellowships (Teaching)

National Science Foundation: GK-12 Fellow—*Full research funding and tuition* 2012-2013
National Science Foundation: GK-12 Fellow—*Full research funding and tuition* 2011-2012
Archambault Teaching Award: Summer@Brown—Second Place 2012
Department of Physics Award for Excellence as a Graduate Teaching Assistant 2010-2011
Brown Presidential Award for Excellence in Teaching: Physics Department Nominee 2010-2011

Committees and Outreach

Lawrence Human Society Volunteer Summer 2017 – Fall 2018
Science teacher: Vartan Gregorian Elementary School 2014 – 2016
Inaugural Graduate student liaison to the curriculum committee: Brown University Physics 2011-2012 2012-2013
Organizer: High Energy Theory Journal Club 2011-2012 2012-2013 2013-2014
Physics department liaison to the Sheridan (teaching) Center 2012-2013 2012-2013 2013-2014
Coordinator of Physics Mentor Program 2012-2013 2013-2014
Coordinator of Physics TA training 2012-2013 2013-2014

Teaching Experience

Quantum Information— PHYS1970F, Course Designer & Instructor, Brown University Spring 2016
Course Grader Brown University—Physics 1600, Computational Physics 2015
Science Teacher Volunteer—4th Grade at Vartan Gregorian Elementary School 2013-2014
Designer & Instructor, Summer@Brown—The Quantum Revolution in Technology 30 Hour High School Course 2012 2013 2014 2015 2016
Course Grader, Brown University—Physics 0070, Advanced Analytic Mechanics 2012 2013 2014
Course Grader, Brown University—Physics 2040, Graduate Electricity & Magnetism 2012
Science Teacher—4th Grade at Vartan Gregorian Elementary School 2011-2012 2012-2013

Community & TA Mentor, Brown University Department of Physics 2011-2016
Science Teacher—Brown Summer High School 2011
Teaching Assistant, Brown University—Physics 0560, Modern Physics 2011
Teaching Assistant, Brown University—Physics 0470, Electricity & Magnetism 2010
Course Grader, University of Michigan—Physics 457, Cosmology 2010
Physics Tutor, University of Michigan Physics Department 2010

Educational Talks/Presentations

"Research in Science"—Graduate Student Panel—SPARK, Brown University 2014 2015
"Effective Teaching Strategies" – Continuing Education Workshop – Brown University May 2013
Effective Scientific Communication Poster Presentation – MRSEC & Continuing Education Brown University June 2013
"What is High Energy Physics?"–Spark Program–Brown Univ. Continuing Education July 2013

Other Projects

In charge of purchase and set-up of High Performance Compute node at University of Kansas
Designing and teaching a new undergraduate/graduate course on "Quantum Information" at Brown University Spring 2016
Helped develop and design a Quantum Entanglement experiment for the instructional labs at Brown University. Students are able to investigate and demonstrate the violation of one of Bell's Inequalities. 2013